

ABHINAV SHARMA

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 Abhinav Sharma |  iabhinavsharma

EDUCATION

- **KIET Group of Institutions**

B.Tech in Electronics and Communication Engineering

◦ CGPA: 7.9/10 (1st Semester)

August 2025 – May 2029 (Expected)

Ghaziabad, India

PROJECTS

- **Accident Detection Helmet**

October 2025

Tools: ESP32, MPU6050, NEO-6M GPS, Arduino IDE, C++



- Wearable helmet device that detects accidents using **ESP32 + MPU6050** and logs GPS location via **NEO-6M** at the moment of impact
- Built a **10-second cancel window** to filter false triggers, making the alert system more reliable in real use
- Wired a battery pack of **3x 650mAh LiPo** cells with a **TP4056** charger and voltage divider for safe, portable power

- **Gesture Controlled Robotic Car**

April 2026

Tools: ESP32, MPU6050, L298N Motor Driver, Bluetooth Classic, Arduino IDE, C++



- **2WD car** that moves based on hand tilt — two **ESP32s** talk over **Bluetooth Classic**, one on the car, one in the controller
- Used **MPU6050** tilt angles to map gestures to directions; added a **dead zone filter** to avoid drift from small hand movements

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, HTML, CSS, JavaScript
- **Microcontrollers & Hardware:** ESP32, Arduino, MPU6050, NEO-6M GPS, L298N Motor Driver, OLED Display, Ultrasonic Sensor, Vibration Sensor, LCD Display
- **Communication Protocols:** Bluetooth Classic, I2C, UART
- **EDA & Design Tools:** EasyEDA, Fusion 360
- **Development Tools:** Arduino IDE, VS Code, Git, GitHub

RELEVANT COURSEWORK

- **Mathematics:** Calculus for Engineers, Linear Algebra for Engineers
- **Electronics & Core:** Digital Electronics, Basic Electrical Engineering, Solid State Physics, Semiconductor Physics & Optoelectronics
- **Programming & Computing:** C Programming, Embedded C, Python, Data Structures, Arduino IDE, IoT Devices and Applications